

VARUN CHOUDHARY

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EDUCATION

University of California, Davis
Bachelor of Science in Computer Science

Davis, CA
Expected June 2027

EXPERIENCE

Co-Founder & Lead Engineer — Lurnigo (EdTech Startup) San Francisco, CA
Leadership & Product Development 2023 – 2025

- Architected the core platform infrastructure helping students pursue top-tier university admissions, scaling the MVP to support personalized guidance workflows.
- Engineered a predictive admission probability engine processing 5,000+ data points (grades, test scores, activity logs) to generate instant acceptance likelihood estimates.
- Led a cross-functional team of 10–20; established CI/CD pipelines and agile sprints to accelerate deployment by 40%.
- Developed Node.js algorithms generating SAT/ACT/GRE learning tracks with real-time analytics progress monitoring.

Software Engineering Intern — SpanIdea Systems Santa Clara, CA
Full-Stack & AI Infrastructure Summer 2025

- Architected a scalable RAG (Retrieval-Augmented Generation) system integrating OpenAI GPT-4 with vector search to serve context-aware responses across Web and Mobile clients.
- Optimized backend latency using FastAPI and Redis caching for repetitive queries while persisting user chat histories.
- Engineered real-time streaming architectures via Server-Sent Events (SSE) to handle concurrent sessions with zero loss.
- Implemented robust rate-limiting and error frameworks to manage token usage, ensuring 99.9% availability during load.

Full Stack Developer — Matellio Hybrid
Frontend & Deployment 2023

- Deployed a high-performance responsive website achieving a 95+ Google Lighthouse score for speed and accessibility.
- Integrated custom analytics and SEO optimizations, directly resulting in a 20–30% increase in inbound customer inquiries.
- Configured SSL, automated CMS workflows, and hosting pipelines to ensure long-term system maintainability.

PROJECTS

Fake News Classification System (NLP — ML)

- Engineered an end-to-end NLP pipeline to detect misinformation, benchmarking TF-IDF baselines against fine-tuned DistilBERT transformers to achieve state-of-the-art classification performance.
- Processed a dataset of 44,898 labeled articles to train robust classifiers, achieving 99.0% test accuracy by optimizing transformer-based embeddings with logistic regression heads.
- Conducted rigorous evaluation using ROC-AUC, F1-score and Precision-Recall curves, performing interpretability analysis to extract influential linguistic features and explain prediction logic.
- Deployed local pipeline returning real-time credibility scores, reducing compute costs by 3x vs BERT models.

Authentication & Access Platform for AI Applications

- Designed a fault-tolerant identity microservice capable of handling 1,000+ concurrent requests, utilizing Redis for sub-millisecond session caching and PostgreSQL for persistent storage.
- Architected secure RESTful endpoints implementing JWT-based session management, role-based access control (RBAC), and automated refresh token rotation policies for seamless access.
- Implemented rate-limiting algorithms to protect compute-intensive AI endpoints from abuse and ensure system stability.
- Enforced OWASP security standards for password hashing and token handling to mitigate SQL injection and XSS risks.

MockMate – AI Voice Interview Coach

- Architected a conversational interview platform leveraging OpenAI's Realtime API to facilitate low-latency voice-to-voice mock interviews, simulating realistic recruiter pressure and timing.
- Engineered a feedback evaluation engine that analyzes audio transcriptions for filler words, tonal confidence, and keyword relevance, generating granular performance scorecards via a React dashboard.
- Implemented bi-directional audio streaming using WebSockets to manage full-duplex communication states, ensuring seamless interruption handling and natural conversation flow between user and AI.

SKILLS

Languages: Python, Java, C++, JavaScript/TypeScript, SQL, Bash
AI & ML: OpenAI API, LangChain, PyTorch, HuggingFace, RAG Pipelines, DistilBERT, NLP, Scikit-learn
Backend & Cloud: Node.js, FastAPI, PostgreSQL, Redis, Docker, AWS, WebSockets, CI/CD
Frontend & Security: React, Next.js, Tailwind CSS, JWT, RBAC, OAuth, OWASP API Security, Linux